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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: R26-DS-010

Project Title: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

1. Introduction
   1.1. Background:

Falls among elderly and mobility impaired patients represent one of the most severe and preventable categories of adverse event in care home and ward settings. The clinical consequences extend beyond the immediate physical injury to encompass prolonged hospitalisation, accelerated functional decline, psychological fear of falling, and, in the most serious cases, patient mortality. From an operational perspective, falls impose significant financial burden on healthcare systems through extended stays, increased nursing workload, and litigation risk.

Existing fall prevention strategies rely on three principal mechanisms: wearable sensor systems, manual nurse observation, and fixed pressure-mat alarms. Each carries a fundamental limitation. Wearable devices depend on continuous patient compliance, which is particularly poor among the confused or agitated elderly populations at highest risk. Nurse observation cannot provide continuous room coverage at the staffing levels typical of modern wards. Pressure mat systems generate high false positive rates that contribute to alarm fatigue a clinically documented phenomenon in which nursing staff begin to habituate to frequent alarms and respond with reduced urgency, or cease responding altogether.

Camera based skeletal pose analysis has emerged as a technically promising alternative: it is non-contact, requires no patient cooperation, and can operate continuously without adding to nursing workload. However, the existing published literature on skeleton-based fall detection suffers from three recurring limitations that restrict clinical adoptability. First, models are typically trained on a single, homogeneous skeleton dataset, making them fragile when deployed with different camera hardware or patient populations. Second, deep learning models optimise for overall classification accuracy but their performance on the clinically critical minority class actual fall events is frequently lower than simpler feature based classifiers, suggesting that complementary model streams exist and are not being exploited. Third, the outputs of existing systems are presented as a binary or probabilistic label with no accompanying explanation, making them clinically unusable in environments where accountability and auditability are mandatory.

This project addresses all three limitations through a unified, edge deployable fall risk detection pipeline that processes live video locally on a consumer laptop, reasons over two complementary skeleton datasets, trains four model architectures evaluated under three escalating difficulty protocols, and presents its predictions through an explainability first web dashboard and caregiver mobile interface.

1.2. Research Problem:

This research addresses the following three-part problem:

- Skeleton format fragility: existing fall-detection models are trained on one skeleton source (either 25-joint Kinect or 18-joint OpenPose) and cannot be applied to a different camera or dataset without retraining from scratch. No published system unifies MediaPipe 33-joint (consumer RGB camera) and Kinect 25-joint (motion-capture) skeletons into one trainable representation.
- Complementary model strengths unexploited: deep models achieve high overall accuracy but consistently underperform simpler hand-crafted feature classifiers on high-risk recall — the most safety-critical metric. Late-fusion architectures that combine both model families are uncommon in the fall-detection literature.
- Lack of clinical explainability: caregivers cannot act on a raw confidence score alone. They require interpretable evidence of why an alert was raised, which joints were unstable, and a brief replay of the leading motion pattern before the event. Current systems provide none of this.

1.3. Objectives:

The table below states the eight primary objectives of this research project, spanning the data pipeline, modelling layer, evaluation rigour, edge deployment, and clinical interface.

IT Number	Objective	Objective Number
IT22057488	Extract skeleton joint coordinates from raw video using MediaPipe Pose and unify them with pre-extracted NTU Kinect skeletons into a common 14-joint representation suitable for multi-dataset fall-risk modelling.	OBJ-1
IT22057488	Design and train a Spatio-Temporal Graph Convolutional Network (ST-GCN) that learns fall-risk patterns directly from raw skeleton sequences by reasoning over joint relationships and temporal motion dynamics simultaneously.	OBJ-2
IT22057488	Engineer 18 clinically interpretable biomechanical motion features from the unified skeleton data (including vertical drop, torso angle, instability score, and body height change) and train a Random Forest baseline and a 4-class Posture Classifier on these features.	OBJ-3
IT22057488	Deploy all processing on a local edge device (consumer laptop, CPU only) with no cloud or GPU dependency, ensuring low latency and patient privacy by design — raw video frames are discarded immediately after pose extraction.	OBJ-4

IT22057488	Evaluate all models rigorously under three escalating evaluation protocols (Cross-Subject, Cross-View, Cross-Dataset) using accuracy, macro F1, and high-risk recall, with a transparent before-and-after data-leakage analysis.	OBJ-5
IT22057488	Combine all model streams into a late-fusion MLP (Fusion MLP) that integrates ST-GCN logits, Random Forest probabilities, and 18 biomechanical features to produce a final 3-class risk decision superior to any individual model.	OBJ-6
IT22057488	Develop a role-based real-time web dashboard (React + Vite + TypeScript) for supervisors and ward managers, displaying live skeletal animations, risk distributions, alert management, event replay, and report generation.	OBJ-7
IT22057488	Integrate the trained backend pipeline with the frontend via a structured JSON manifest, so every risk score, probability, posture label, and feature attribution shown in the dashboard originates from actual model inference on held-out test sequences.	OBJ-8

2. Data Exploration

2.1. Data Collection

Two publicly available datasets were selected to provide mutually complementary coverage of the problem. The UR Fall Detection Dataset provides real-world deployment realism: RGB video frames captured by consumer cameras, processed through MediaPipe Pose to extract skeleton data, with realistic noise from partial occlusion, side-view geometry, and variable lighting. The NTU RGB+D 60 dataset (fall-related subset) provides statistical volume and demographic diversity: pre-extracted Kinect skeleton data from 106 performers across 3 camera viewpoints, free of pose-extraction noise. Neither dataset alone satisfies all eight research objectives; their combination is a deliberate design decision.

For the UR dataset, each PNG image sequence was processed frame-by-frame through MediaPipe Pose (model_complexity=1, min_detection_confidence=0.5), extracting 33 landmark coordinates (x, y, z, visibility) per frame and saving the result as a (T, 33, 4) NumPy array. For the NTU dataset, each .skeleton binary file was parsed using a custom extractor (ntu_skeleton_extract.py) to yield the primary detected body's 25 joint coordinates as a (T, 25, 3) array. Both outputs were unified through the 11-step preprocessing pipeline.

2.2. Dataset Description:

Data Source	Description	Resource	Size	Key Attributes
UR Fall Detection Dataset	RGB image sequences of real falls and daily activities captured with consumer cameras at side and overhead angles. The only publicly available fall dataset using real RGB video with genuine fall events.	University of Rzeszow (public, academic use)	60 raw sequences; 45 after quality filtering (30 falls + 15 ADL)	PNG frames per sequence; 2 camera angles (cam0 overhead, cam1 side); fall and ADL labels; MediaPipe 33 landmarks x (x,y,z,visibility)
NTU RGB+D 60 (Fall-Related Subset)	Large-scale pre-extracted Kinect skeleton dataset. Five fall-related action classes selected: sit_down (A008), stand_up (A009), staggering (A042), falling_down (A043), squat_down (A080).	ROSE Lab, Nanyang Technological University (public, academic use)	4,752 sequences; 106 performers; 3 camera setups (C001, C002, C003); 32 sessions	.skeleton binary files; 25 Kinect joints x (x,y,z); action codes A008, A009, A042, A043, A080; subject-level performer metadata

NTU action-to-risk-level mapping applied during data extraction:

i. A008 (sit_down) and A009 (stand_up) -> Low Risk: controlled, voluntary posture transitions with no balance-loss component.
ii. A042 (staggering) and A080 (squat_down) -> Moderate Risk: ambulatory instability or deep low-centre-of-gravity postures linked to pre-fall conditions.
iii. A043 (falling_down) -> High Risk: full uncontrolled body collapse to the floor.
iv. UR fall sequences (cam0 + cam1) -> High Risk; UR ADL sequences -> Low Risk (normal daily activities).

Class distribution in the combined dataset (4,797 total sequences):

Risk Class	Approx. Count (Test Set, n=799)	Source Action Labels	Class Handling Strategy
Low Risk	~326 (41%)	NTU A008, A009; UR ADL (normal)	class_weight='balanced' on RF; class-weighted cross-entropy loss on ST-GCN and Fusion MLP
Moderate Risk	~306 (38%)	NTU A042, A080	Same balanced weighting applied across all models
High Risk	~167 (21%)	NTU A043; UR fall (cam0 + cam1)	Minority but clinically most critical class. High-risk recall is the primary safety metric reported throughout.

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

The table below explains how well each dataset aligns with the research problem and each of the eight objectives. OBJ-4 (edge deployment) and OBJ-6 (fusion) are not separately listed as they draw on data produced by the other objectives.

	OBJ-1: Skeleton Extraction	OBJ-2: ST-GCN Training	OBJ-3: Feature Engineering	OBJ-5: Evaluation Protocols	OBJ-7/8: Dashboard & Integration
UR Fall Detection Dataset	High. RGB frames processed through MediaPipe frame-by-frame; 33 landmarks per frame with carry-forward imputation on failed detections.	Partial. 45 sequences alone are insufficient to train a 3M-parameter ST-GCN. Used combined with NTU under the Cross-Subject protocol.	High. RGB-derived skeletons introduce realistic occlusion noise and side-view geometry challenges that clean motion-capture data cannot replicate.	High. Provides the real-world domain target for Cross-Dataset evaluation (CD-NTU->UR), stress-testing lab-to-real-camera generalisation.	Partial. Too small for standalone quantitative evaluation but real model predictions on UR test sequences populate some patient slots in the dashboard.
NTU RGB+D 60 (Fall Subset)	High. Pre-extracted .skeleton files parsed directly to (T,25,3)	High. 4,752 sequences across 106 performers provide	High. Clean Kinect skeletons enable precise computation of all 18	High. Rich subject-level and camera-level metadata supports all	High. 24 patient-slot predictions in the PP1 dashboard are

	NumPy arrays; no video processing required.	sufficient volume and diversity for ST-GCN convergence without overfitting.	biomechanical features without occlusion-induced noise distortion.	three evaluation protocols. Cross-View (C001 vs C002+C003) is NTU-exclusive.	drawn from NTU held-out test sequences with real model outputs shown per patient.

Scalability assessment: The UR dataset (45 sequences) is sufficient for quality-filtered pose extraction and as a real-world held-out evaluation target but is intentionally not used as a standalone training corpus for deep models. The NTU subset (4,752 sequences, 106 performers, 3 cameras) provides the scale and diversity required to train a 3-million-parameter deep model and to support all three evaluation protocols independently. Together, 4,797 unified (100, 14, 3) sequences form the training corpus. Under the Cross-Subject split, this yields 3,121 training, 877 validation, and 799 test sequences a statistically stable partition. After preprocessing, raw data totalling ~12 GB is reduced to under 100 MB of NumPy arrays and feature CSVs, making the system fully edge deployable.

Demographic and generalisability limitations: both datasets feature young adult performers, not the elderly or cognitively impaired patients who represent the highest-risk clinical population. NTU skeletons are geometrically cleaner than MediaPipe extractions from UR consumer-camera footage, creating a modality gap that the Cross-Dataset protocol explicitly quantifies. These limitations are addressed prospectively through the planned Phase 3 in-house laboratory study with elderly volunteers using the actual deployment camera hardware (Hikvision DS-U02, wall-mounted at 6-7 feet).

3. Methodology

3.1. Data Preprocessing:

The preprocessing pipeline transforms raw, heterogeneous source data from two incompatible skeleton formats into a single unified (100, 14, 3) tensor shared by all downstream models. The pipeline comprises 11 sequential steps; four steps marked with [NOVEL] are original contributions not present in the reviewed reference literature (Yan et al. 2018; Shi et al. 2019; Rehouma & Boukadoum 2024).

Techniques applied from the standard preprocessing toolkit include data cleaning and quality filtering, temporal normalisation, dimensionality reduction, feature engineering, data integration across heterogeneous schemas, spatial normalisation, handling of missing detection data, data type conversion, outlier/weak-sequence removal, and subject-level data splitting with leakage assertion.

Objective 1 — Skeleton Extraction and Unification (UR Dataset and NTU Dataset)

The following transformation techniques were applied per dataset for Objective 1:

Dataset	Transformation Technique Applied
UR Dataset	MediaPipe Pose extraction (model_complexity=1, min_detection_confidence=0.5) applied frame-by-frame to PNG image sequences. Extracts 33 landmarks x (x, y, z, visibility) per frame. Output: (T, 33, 4) NumPy array per sequence. Script: ur_pose_extract.py:51-101.
UR Dataset	**[NOVEL]** Carry-forward missing-pose imputation: when MediaPipe fails on a frame due to occlusion or blur, the last valid pose is copied forward instead of inserting all-zero rows. This prevents the model from learning "zeros = fall" as a spurious artefact. Script: ur_pose_extract.py:74, 96-97.
UR Dataset	Sanity validation: checks file existence, array shape integrity, and flags sequences shorter than 20 frames. Script: check_pose_output.py:42-78.
UR Dataset	**[NOVEL]** Quality-tier filtering: each sequence is classified as Excellent (missing_ratio = 0.00), Good (<=0.20), Acceptable (<=0.40), or Weak (>0.40). Weak sequences are removed from training. UR reduced from 60 to 45 usable sequences. Script: filter_weak_sequences.py:24, 52-59.
NTU Dataset	Binary .skeleton file parsing: reads each frame's 25 joint (x, y, z) coordinates for the first detected body. Script: ntu_skeleton_extract.py:52-121.
NTU Dataset	Action-to-risk-level mapping: A008/A009 -> Low Risk; A042/A080 -> Moderate Risk; A043 -> High Risk. UR falls -> High Risk; UR ADL -> Low Risk. Script: ntu_skeleton_extract.py:21-42.
NTU Dataset	Shape validation: flags missing skeleton files, invalid array dimensions, and sequences shorter than 20 frames. Script: check_ntu_output.py.

Both Datasets	Temporal normalisation to T=100 frames: long sequences are evenly subsampled using np.linspace indexing. Short sequences use **[NOVEL]** linear-interpolation padding (replaces iteration-1 frame-freeze, which created a static-tail artefact correlated with the High-Risk label). Script: normalize_sequences.py:31-56.
Both Datasets	**[NOVEL]** Cross-schema 14-joint mapping: 13 anatomical joints common to both MediaPipe-33 (UR) and NTU-25 are selected via index lookup. A 14th synthetic spine joint is computed as the mean of the left/right shoulder midpoint and left/right hip midpoint. Output shape: (100, 14, 3). Script: map_common_joints.py:24-95.
Both Datasets	**[NOVEL]** Spine-centred, torso-scaled spatial normalisation: (a) the spine joint is translated to the origin per frame, removing absolute position as a predictive shortcut. (b) all coordinates are divided by the mean torso length (spine to shoulder-midpoint), normalising for body size differences between performers. Script: map_common_joints.py:98-125.
Both Datasets	Subject-level disjoint split with leakage assertion: NTU performers (P001-P106) and UR session IDs (adl/fall-NN) are assigned exclusively to train, validation, or test sets. A code-level assertion verifies disjoint before training begins. Script: cross_subject_split.py:75-167.

Objective 2 — ST-GCN Training Data Preparation (Both Datasets Combined)

Dataset	Transformation Technique Applied
Both (Combined)	Input to ST-GCN: unified (100, 14, 3) skeleton arrays from the Common Joint Sequences output directory. The same arrays produced by OBJ-1 preprocessing are used directly, ensuring no intermediate processing introduces discrepancies.
Training Only	**[NOVEL]** Training-time data augmentation (ST-GCN only; disabled for validation, test, and inference): (a) random rotation +-15 degrees around the vertical Y-axis; (b) horizontal mirror with corresponding left/right joint-index swap; (c) time-jitter: five randomly selected frames are dropped and the sequence is re-interpolated back to 100 frames. Each transform fires with probability 0.5 per sample. Script: stgcn/dataset.py.
All Protocols	Three evaluation protocol splits applied from the same preprocessed data: (1) Cross-Subject: cross_subject_split.py -> splits/ [3,121/877/799 sequences]; (2) Cross-View: cross_view_split.py -> splits_cv/ [2,596/572/1,584]; (3) Cross-Dataset NTU->UR: cross_dataset_ntu2ur_split.py -> splits_cd_ntu2ur/ [3,894 NTU train / 858 NTU val / 45 UR test]. Protocol switched via MODELS_PROTOCOL environment variable (cs / cv / cd_ntu2ur).

Objective 3 — Feature Engineering and Classical Model Preprocessing (Both Datasets)

Dataset	Transformation Technique Applied
Both (Combined)	Motion feature extraction from unified (100, 14, 3) sequences: each sequence is reduced to 18 scalar biomechanical features using finite-difference kinematics, trigonometric torso-angle computation, and Euclidean body-height measurement. Output: features_dataset.csv (4,797 rows x 23 columns). Script: extract_motion_features.py.
Both (Combined)	Dimensionality reduction: 100 x 14 x 3 = 4,200 numbers per sequence compressed to 18 interpretable scalars (a 233x reduction). Features are coordinate-invariant by construction and require no additional normalisation for Random Forest or Posture RF inference.
NTU (Labels)	Posture label derivation: NTU action codes mapped to four posture classes. A043 and UR falls -> Lying; A008 and A080 -> Sitting; A009 -> Standing; A042 -> Walking. UR ADL sequences excluded from posture training (mixed postures within a single clip, ambiguous labelling). Script: posture_rf.py.

Objectives 4, 5 and 6 — Edge Deployment, Evaluation, and Fusion Preprocessing

Objective	Transformation Technique Applied
OBJ-4: Edge	No additional preprocessing required. All model files (RF pkl ~8MB, ST-GCN pt ~12MB, Fusion pt ~15KB) are saved to disk after training and loaded at inference time on the consumer CPU. Raw video frames are discarded immediately after MediaPipe extraction only the anonymous 14-joint skeleton is ever stored.
OBJ-5: Evaluation	Comparison report aggregation: generate_comparison_report.py reads every (model x protocol) result CSV and combines all 12 checkpoints into comparison_all.csv, comparison_per_dataset.csv, and comparison_summary.md. Before/after leakage proof documented: Iteration-1 random split RF = 92.78%; Iteration-2 subject-disjoint RF = 87.48% (Delta = -5.3%).
OBJ-6: Fusion	Late-fusion 24-dimensional input assembled per sequence: ST-GCN softmax logits (3 values) + Random Forest predict_proba outputs (3 values) + 18 motion features = 24 scalars. No new preprocessing; values drawn from existing model inference outputs and features_dataset.csv. Script: fusion_mlp.py.

Objectives 7 and 8 — Dashboard Integration and Backend-Frontend Data Pipeline

Component	Transformation Technique Applied
Backend -> Manifest	build_demo_predictions.py reads held-out test predictions from all four model outputs (test_predictions.csv for ST-GCN and Fusion MLP; classification reports for RF and Posture RF), selects 24 unique test sequences matching the dashboard patient risk distribution (5 High, 8 Moderate, 11 Low), and writes a structured JSON manifest. All values originate from actual model inference.
Skeleton Rendering	skeleton_renderer.py reads each assigned sequence's (100, 14, 3) .npy file, auto-detects Y-axis orientation (UR inverted, NTU upright), and renders a 24 fps MP4 stick-figure animation for each of the 24 patient slots. These videos are embedded in the dashboard's Live View panel, showing supervisors the exact skeleton the model received.
Explainability Generation	build_justification() generates a plain-English explanation per patient citing: the action annotation, the five most discriminative motion feature values (from features_dataset.csv), and a templated reasoning sentence based on the predicted risk class. The explanation is faithful to what the model actually used — it is not a post-hoc rationalisation.
Frontend Data Contract	The manifest.json schema includes: patientId, room, source dataset, sequenceId, actionLabel, riskLevel, riskScore, posture, confidence, probability vectors (pLow/pModerate/pHigh for Fusion; rfLow/rfMod/rfHigh for RF; stgcnLow/stgcnMod/stgcnHigh for ST-GCN), groundTruth, correct flag, justification string, and a features object with 8 key motion metric values. The frontend reads this at app launch via demoFeed.ts.

3.2. Scalability

The UR dataset (45 sequences) is suitable for pose extraction and real-world domain evaluation, but insufficient as a standalone training corpus for deep models. The NTU subset (4,752 sequences, 106 performers, 3 cameras) is highly scalable for deep model training and supports all three evaluation protocols independently. The combined 4,797-sequence corpus yields a statistically stable 70/15/15 subject-disjoint split (3,121/877/799). After preprocessing, raw data totalling approximately 12 GB is reduced to under 100 MB of NumPy arrays and CSVs, making the full pipeline edge-deployable on a consumer laptop. The trained model suite totals approximately 21 MB (RF pkl + ST-GCN checkpoint + Fusion checkpoint combined).

3.3. Feature extraction

The 18 biomechanical motion features are extracted from each unified (100, 14, 3) skeleton sequence. They constitute the sole input to the Random Forest and Posture Classifier, and one of three input streams to the Fusion MLP. Each feature group is described in the table below, along with the clinical relevance that motivated its inclusion.

Feature Group	Feature Names	Clinical Relevance
Centre-body kinematics	center_speed_mean, center_speed_max, center_speed_std	Mean position of all 14 joints per frame; speed derived by first-order finite differences. Falls involve a characteristic acceleration spike followed by abrupt deceleration on impact.
Centre-body acceleration	center_acceleration_mean, center_acceleration_max, center_acceleration_std	Rate of velocity change. High, sudden acceleration distinguishes loss of control from smooth voluntary movement such as sitting down or reaching.
Vertical movement	vertical_range, vertical_drop, sudden_vertical_change	Captures downward excursion of the body centre along the Y-axis. vertical_drop measures maximum single-direction drop; sudden_vertical_change flags abrupt frame-to-frame displacement — the primary fall signal.
Torso angle and body tilt	torso_angle_mean, torso_angle_max, torso_angle_std	Angle of the torso vector (shoulder-centre to hip-centre) relative to vertical, computed per frame. High tilt and high variance indicate instability and are associated with pre-fall postural sway.
Body height	body_height_change, body_height_range	Euclidean distance from head to ankle-centre per frame, normalised by torso length. A standing-to-lying transition produces near-complete collapse in body height — the most discriminative postural signal.
Per-joint speed	mean_joint_speed, max_joint_speed, std_joint_speed	Average, peak, and standard deviation of per-joint frame-to-frame displacement across all 14 joints. Captures global motion intensity and joint-specific instability across the sequence.
Instability composite	instability_score	Custom scalar combining body-centre lateral sway (oscillation of spine joint) with per-joint speed variability. Designed to mirror clinical assessment of balance impairment in physiotherapy settings.

Joint mapping — the 14-joint common schema resolves the MediaPipe-33 vs NTU-25 incompatibility:

Common Joint Name	MediaPipe-33 Index (UR)	NTU-25 Index, 0-based (NTU)
Head	0 (nose)	3
Left Shoulder	11	4
Right Shoulder	12	8
Left Elbow	13	5
Right Elbow	14	9
Left Wrist	15	6
Right Wrist	16	10
Left Hip	23	12
Right Hip	24	16
Left Knee	25	13
Right Knee	26	17
Left Ankle	27	14
Right Ankle	28	18
Spine (synthetic, computed)	Average of shoulders + hips	Average of shoulders + hips

4. Modelling and Results

The results section presents the key findings derived from training and evaluating four model architectures under three evaluation protocols, producing 12 trained model checkpoints in total (4 architectures x 3 protocols).

4.1. Key Insights:

Four model architectures were built and evaluated. Each is described below with its input, output, design rationale, and headline performance figures.

Model 1 — Random Forest Baseline: Input: 18 biomechanical motion features (scalars). Output: 3-class risk (Low / Moderate / High). An ensemble of 300 decision trees trained on the 18 hand-crafted features. It serves three simultaneous roles: a quantitative baseline for comparison against the deep ST-GCN; a lightweight edge fallback (~8 MB, sub-millisecond inference) for real-time use on consumer hardware; and a source of Gini importance rankings that directly power the explainability heatmap shown to clinical users, identifying which biomechanical signals drove each prediction. class_weight="balanced" applied to counteract the 2:2:1 class imbalance. Five-fold stratified cross-validation on training set only; variance +-0.68%, confirming stability.

Cross-Subject test accuracy: 87.48%; macro F1: 86.18%; high-risk recall: 74.85%.

Model 2 — Posture Classifier (Random Forest): Input: 8 posture-specific features (torso inclination, knee-to-hip drop ratio, ankle motion, body height statistics). Output: 4-class posture label (Lying / Sitting / Standing / Walking). Addresses a distinct clinical question from the risk classifier: what body position the patient is currently in. Posture context modulates clinical interpretation of a risk score — a high-risk score from a standing patient warrants a different response than the same score from a patient who is already lying down. Labels are derived from NTU action codes (action-derived, not geometric heuristics). UR ADL sequences excluded due to mixed postures within single clips.

Cross-Subject test accuracy: 74.15%; macro F1: 74.71%.

Model 3 — ST-GCN (Spatio-Temporal Graph Convolutional Network): Input: raw unified skeleton sequences (100 frames x 14 joints x 3 coordinates). Output: 3-class risk. The primary deep-learning model. Unlike the Random Forest, which summarises motion statistics, the ST-GCN learns spatio-temporal motion patterns directly from raw joint trajectories. It treats the skeleton as a graph (joints = nodes, bones = edges), applying graph convolution across space and temporal convolution across the 100-frame sequence simultaneously, capturing how the motion unfolds over time. Architecture: 7 ST-GCN blocks (spatial graph conv -> temporal conv k=9 -> BatchNorm -> ReLU -> residual) + global average pooling + linear head. 3,071,247 trainable parameters. The custom 14-joint anatomical adjacency graph is an original design of this project. Trained CPU-only (HP Victus 15); ~151 minutes per Cross-Subject run. Cross-Subject test accuracy: 91.86%; macro F1: 90.71%; high-risk recall: 82.63%.

Model 4 — Fusion MLP (Late-Fusion Architecture): Input: 24-dimensional vector: ST-GCN logits (3) + Random Forest predict_proba (3) + 18 biomechanical features. Output: 3-class risk (final system prediction displayed in the dashboard and mobile app). The central contribution of this research. Motivated by the empirical observation that the RF and ST-GCN exhibit a complementary recall-precision trade-off: RF achieves higher high-risk recall; ST-GCN achieves higher overall accuracy. The Fusion MLP learns to route predictions through whichever model stream is more reliable for a given input.

Architecture: Linear(24->64) -> BatchNorm -> ReLU -> Dropout(0.3) -> Linear(64->32) -> BatchNorm -> ReLU -> Dropout(0.2) -> Linear(32->3). Only 3,811 parameters; trains in ~2 minutes on CPU. Cross-Subject test accuracy: 94.24%; macro F1: 93.54%; high-risk recall: 87.43%.

Cross-View test accuracy: 94.07%; high-risk recall: 90.19%. Best in-domain model across all metrics.

Full results matrix across all 12 checkpoints (4 architectures x 3 evaluation protocols):

Model	Protocol	Accuracy	Macro F1	High-Risk Recall	Missed Falls (approx.)
RandomForest Baseline	Cross-Subject	87.48%	86.18%	74.85%	~25%
RandomForest Baseline	Cross-View	90.91%	89.95%	82.91%	~17%
RandomForest Baseline	Cross-Dataset NTU->UR	20.00%	15.00%	0.00%	N/A
Posture RF (4-class)	Cross-Subject	74.15%	74.71%	--	--
Posture RF (4-class)	Cross-View	77.97%	78.72%	--	--
Posture RF (4-class)	Cross-Dataset NTU->UR	23.33%	9.46%	--	--
ST-GCN	Cross-Subject	91.86%	90.71%	82.63%	~17%
ST-GCN	Cross-View	88.89%	87.67%	88.29%	~12%
ST-GCN	Cross-Dataset NTU->UR	4.44%	5.56%	0.00%	N/A
Fusion MLP (BEST)	Cross-Subject	94.24%	93.54%	87.43%	~12%
Fusion MLP (BEST)	Cross-View	94.07%	93.46%	90.19%	~10%
Fusion MLP	Cross-Dataset NTU->UR	0.00%	0.00%	0.00%	N/A

Top-5 features by Random Forest Gini importance (Cross-Subject): (1) vertical_range (0.203) — total vertical excursion; (2) vertical_drop (0.162) — maximum single downward displacement; (3) center_speed_mean (0.122) — mean body-centre velocity; (4) body_height_change (0.116) — standing-to-lying height collapse; (5) mean_joint_speed (0.075) — global joint motion intensity. These rankings align with clinical biomechanical domain knowledge, validating that the model reasons correctly rather than exploiting statistical artefacts.

Cross-Dataset domain shift: The Cross-Dataset protocol (train on NTU, test on UR) reveals a systematic collapse across all architectures (RF 20%, Posture RF 23%, ST-GCN 4%, Fusion MLP 0%). This is a substantive research finding, not a failure to hide. The ST-GCN achieves 96.10% training and 96.39% validation accuracy on NTU source domain, confirming that the collapse is domain shift (Kinect vs MediaPipe skeleton modality mismatch) rather than overfitting. This motivates: (1) combined NTU+UR training as essential for any deployment-ready model; (2) the Phase 3 in-house lab study with deployment-domain cameras as a load-bearing research step; and (3) per-dataset metrics reported alongside headline accuracy to prevent misleading summaries.

Overfitting and leakage defences — seven independent lines of evidence confirm results are not inflated:

i. One-touch test set: test split evaluated exactly once inside evaluate.py, after training is completely frozen. Best checkpoint selected exclusively on the validation split.
ii. Subject-disjoint split with code assertion: NTU performers (P001-P106) and UR session IDs assigned exclusively to one partition. An explicit assertion verifies disjointness before training proceeds.
iii. 5-fold CV variance on RF training set: +-0.68% — signature of a generalising model, not a memorising one.
iv. Regularisation hygiene on all deep models: early stopping (patience 10), Dropout (0.3/0.2), BatchNorm, weight decay (1e-4), class weighting, and training-time augmentation.
v. Training curve integrity: ST-GCN best epoch was 27 of 37 (Cross-Subject) — not the final epoch, confirming early stopping identified genuine convergence.
vi. Task difficulty calibration: 3-class risk classification is structurally easier than 60-class NTU action recognition. ST-GCN's 81.5% on 60-class NTU scales correctly to expected 90%+ on a 3-class task.
vii. Before/after leakage proof: Iteration-1 random split RF = 92.78%; Iteration-2 subject-disjoint RF = 87.48%. Delta = -5.3% confirms the subject-disjoint constraint removed genuine leakage.

Objectives 7 and 8 — Dashboard and Integration Results: The React + Vite supervisor dashboard is fully implemented and operational as of Phase 1. It provides six functional tabs: Dashboard (24-patient overview with live risk scores and skeletal animations), Room Overview (bed-level occupancy and risk

distribution per ward), Alerts and Risk (acknowledgement and resolution workflow with audit trail), Event Replay (skeletal animation playback for historical events), Reports (PDF and CSV generation), and Settings (admin-only configuration). The sustained moderate alert mechanism fires only after a patient's risk score remains in the moderate band for approximately 12 continuous seconds, eliminating noise from brief fluctuations. Pose-quality gating suppresses inference for patients with degraded or unavailable skeleton data. Every value displayed in the dashboard — risk scores, probability vectors, posture labels, feature attributions — originates from actual trained model inference on held-out test sequences via the JSON manifest integration (OBJ-8).

4.2. Challenges Faced During Data Analysis:

Challenge	Description	Resolution Applied
Incompatible skeleton schemas	MediaPipe outputs 33 joints; Kinect outputs 25 joints. No shared joint format existed for unified training across the two datasets.	Designed a 14-joint common anatomical schema; added a computed synthetic spine joint (mean of shoulders + hips). Implemented in map_common_joints.py.
MediaPipe detection failures on UR	Partial occlusion on side-view falls and floor-proximity poses caused entire frames to fail detection, generating all-zero rows that the model could learn as artefacts correlated with high-risk clips.	Carry-forward imputation: last valid pose reused on failed frames. Sequences with missing_ratio > 0.40 removed entirely as Weak-quality.
Cross-dataset coordinate shift	NTU and UR skeletons occupy different metric spaces and camera perspectives. Models trained on NTU could memorise absolute coordinate ranges as a dataset-identity shortcut rather than learning motion patterns.	Spine-centred, torso-scaled spatial normalisation applied per frame. Confirmed effective: Iteration-2 CS accuracy dropped 5.3% relative to Iteration-1, proving the shortcut was removed.
Class imbalance (2:2:1)	High-risk class represents only 21% of the data but is the clinically most important class. Without correction, classifiers bias toward majority classes and underperform on high-risk recall.	class_weight="balanced" for RF and Posture RF; class-weighted cross-entropy loss for ST-GCN and Fusion MLP. High-risk recall adopted as primary safety metric throughout.
UR dataset size (45 sequences)	45 sequences are insufficient to train a 3M-parameter ST-GCN without severe overfitting, as the model capacity far exceeds the available training samples.	UR combined with NTU for Cross-Subject training (4,797 sequences combined). UR used as a standalone held-out domain for Cross-Dataset evaluation only.

Person-identity leakage (Iteration 1)	Random stratified splits placed the same performer in both train and test sets, inflating CS accuracy by approximately 5% across all models. The system was inadvertently learning person identity, not motion patterns.	Replaced with subject-disjoint splits using NTU performer codes (P###) and UR session IDs. Code assertion verifies correctness. Full Iteration-1 vs Iteration-2 comparison reported as a leakage proof.
CPU-only training constraint	No GPU available. ST-GCN (3M parameters, 100-frame inputs, batch size 32) required approximately 151 minutes per Cross-Subject training run on the consumer laptop CPU.	Architecture sized for CPU feasibility (7 blocks); cosine LR annealing over 50 epochs; early stopping applied to avoid unnecessary additional epochs beyond convergence.
Frame-freeze padding artefact (Iteration 1)	Iteration-1 padded short sequences by repeating the last frame. Fall sequences end with the body on the floor, so N identical floor-pose tail frames created a detectable static-tail signal correlated with the High-Risk label.	Replaced with linear interpolation across the full sequence length. Tail artefact eliminated and confirmed by accuracy reduction in Iteration-2 evaluation.
Cross-dataset domain shift (CD protocol)	All deep models collapse to 0-25% accuracy when trained on NTU and tested on UR, even after spatial normalisation. Domain shift between Kinect and MediaPipe skeleton modalities is severe.	Reported honestly as a research finding rather than suppressed. Motivates combined NTU+UR training, the planned Phase 3 in-house lab study with deployment-domain cameras, and mandatory per-dataset metric reporting.

5. AI USAGE DISCLOSURE

AI tools were used in a limited and supportive capacity during the preparation of this data analysis report. ChatGPT ,Google Gemini and Claude was utilized primarily to enhance language clarity, refine sentence structure, and assist in the organization of ideas. The purpose of using these tools was to improve the readability and presentation of the document, while ensuring that all core research concepts, problem formulation, and methodological decisions were developed independently by the author.

The extent of AI usage was strictly limited to minor writing assistance and idea refinement and did not involve generating full sections of the proposal or replacing original intellectual contribution. All AI-assisted outputs were critically reviewed, modified, and validated by the author to ensure accuracy, relevance, and alignment with academic standards. Validation methods included cross-checking information with credible academic sources, verifying technical correctness, and ensuring consistency with the research objectives. Full responsibility for the content presented in this proposal is retained by the author.

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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